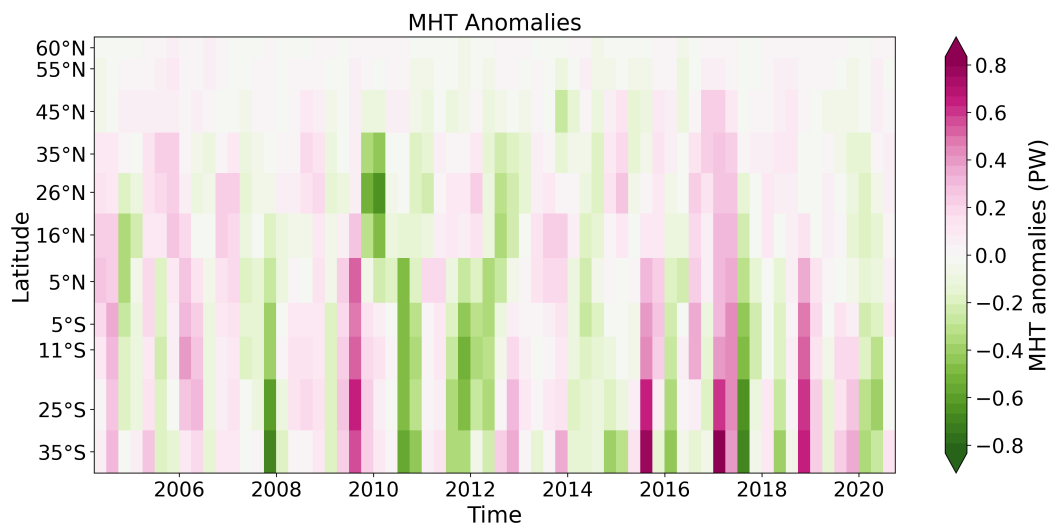


Masterthesis

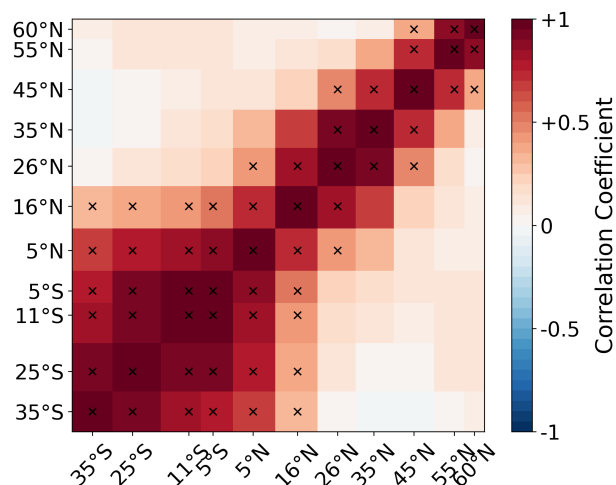
1. Concept - Abstract

1.1 Results:

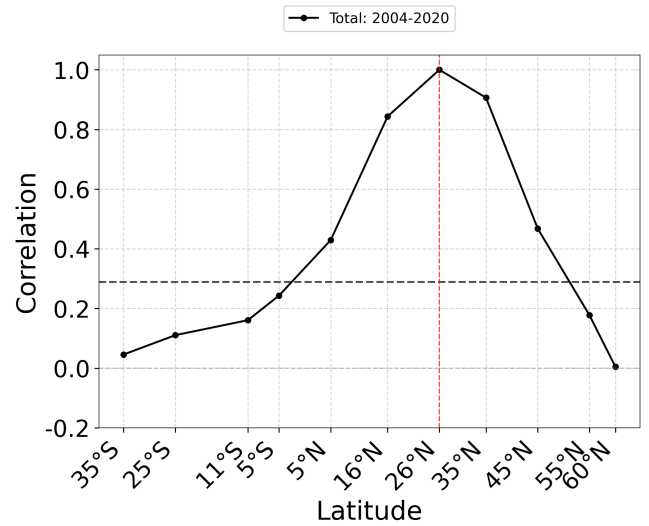
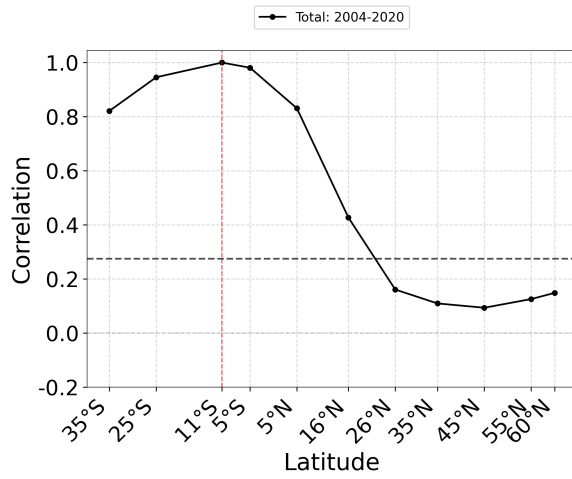
1. The hovmöller diagram of the meridional heat transport anomalies hint towards latitudes where anomalies vary together, namely between 35°N - 16°N and 5°N - 35°S . Overall weakest anomalies north of 35°N , in subpolar region.



2. The cross-correlation between all latitudes confirms that these latitude bands tend to vary together, which can be seen from the high pairwise correlation between those.

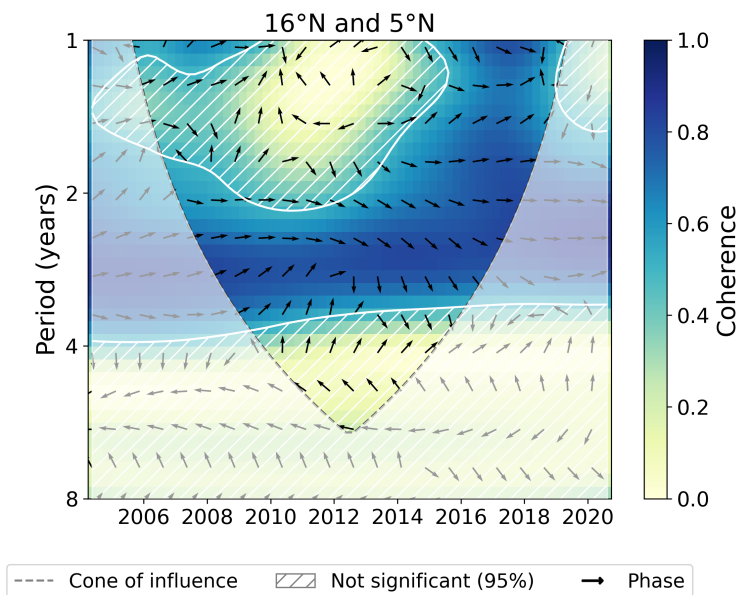


3. The 0-lag correlation for 11°S with all latitude bands confirms that the southern latitudes vary together, and the relationship drops north of 5°N . Similarly, the latitudes in the subtropical north Atlantic show a high correlation in itself, with a stark drop in the connection south of 16°N and north of 35°N - maybe not needed since its just the same finding as 2.



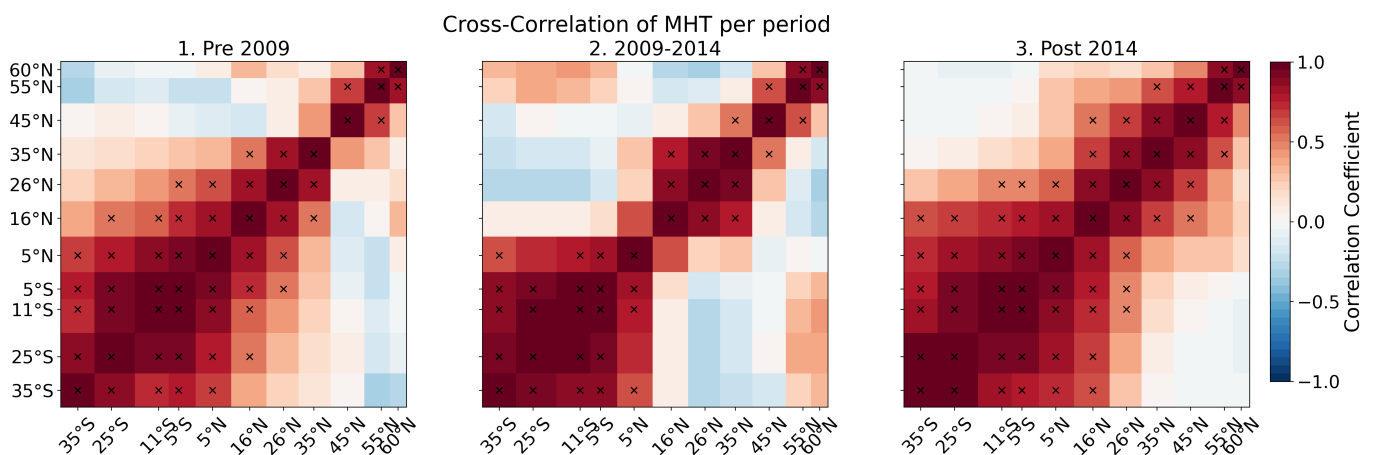
4. The wavelet coherence between 16 and 5°N reveals a drop in the coherence of the meridional heat transport occurring between 2009-2014.

4.1 Three periods can be distinguished: 2004-2008, 2009-2014, 2015-2020



5. MAIN FIGURE

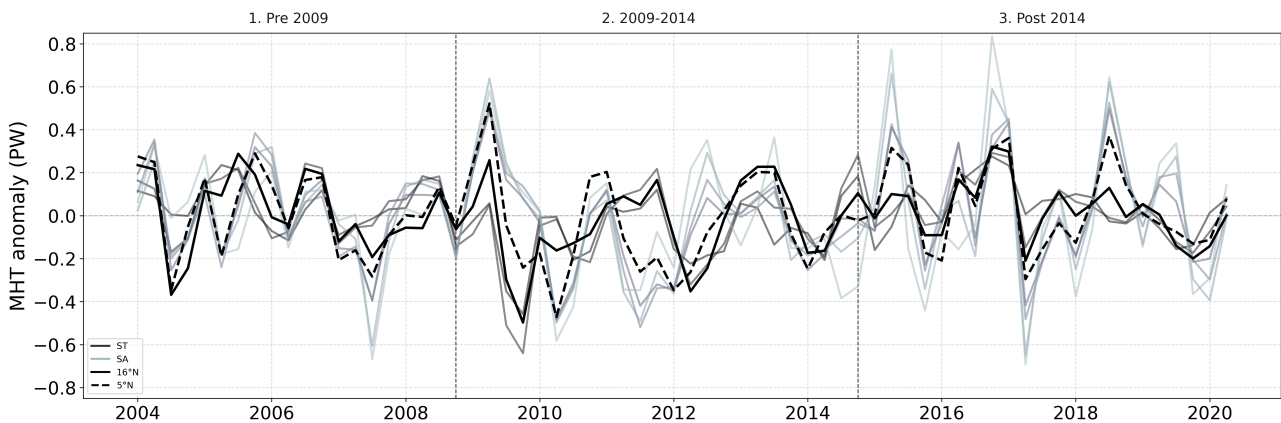
The cross correlation done in 1. Redone for the 3 periods reveals that prior to 2009 and after 2014 the meridional heat transport seems coherent throughout the atlantic south of 35°N, while in 2009-2015 the coherence splits, leaving two boxes of coherent MHT.



6. The average correlation between the two blocks drops for the indicated period from 0.53 to a no longer significant value.

	A	B	C
1	Period	Corr ST & SA block	Significant (95%)
2	Pre 2009	0.53	Yes
3	2009-2014	0.02	No
4	Post 2014	0.47	No (rcrit: 0.48)

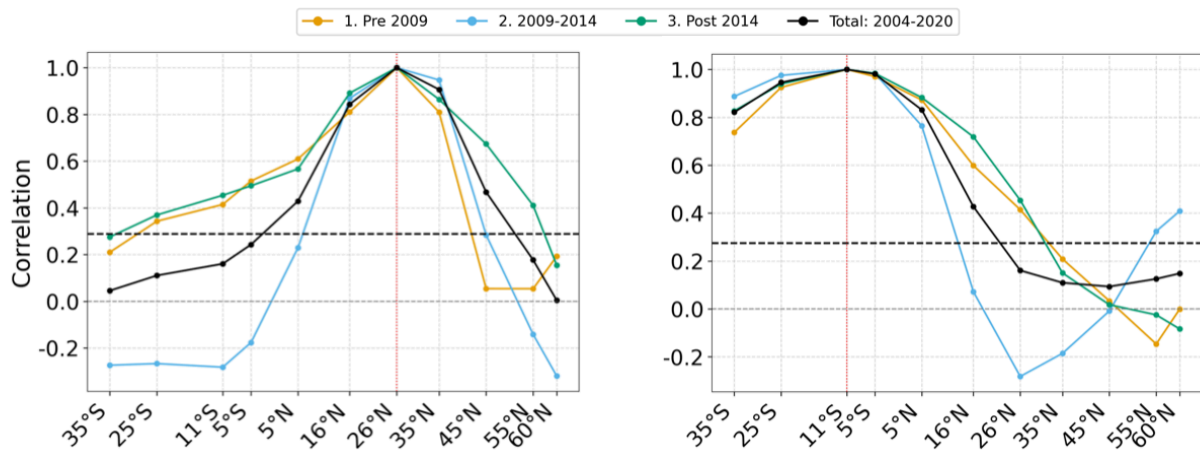
(At the beginning of the 2nd period the subtropical north atlantic latitudes anomalies show the opposite behavior of the south atlantic ones.) ? Idea to show the timeseries but unsure about what to say



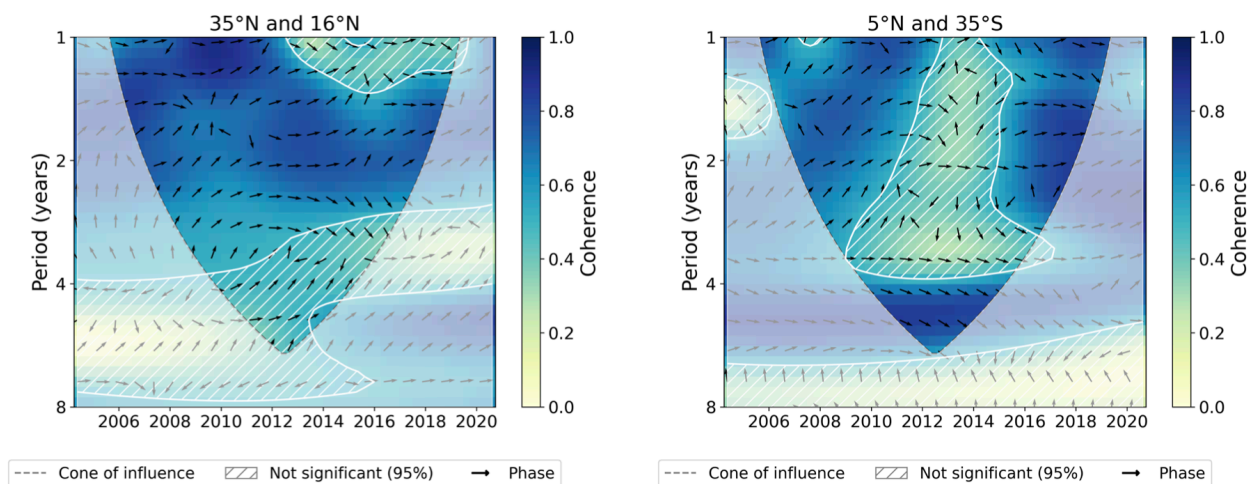
7. The blocks in itself stay coherent throughout the whole timeseries, with just slight changes in the connection.

(can show this either with 0-lag correlation or wavelet coherence)

0-lag again 26°N and 11°S



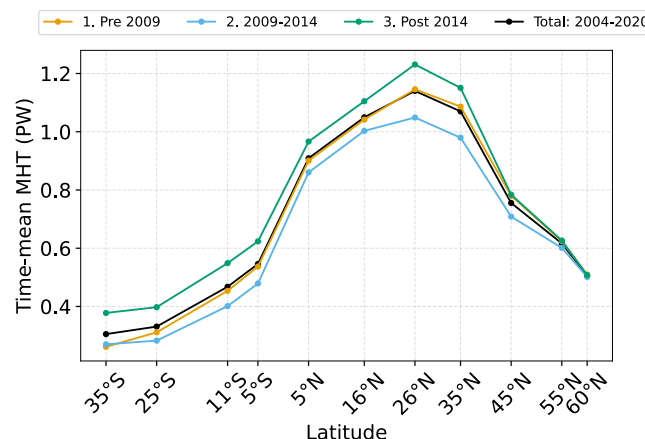
Or WCT: Subtropical north atlantic & south atlantic latitudes:



8. Compared to the pre2009 baseline period, the MHT reduces at all latitudes, with the strongest reduction at 26°N. After 2014 the MHT recovers, while even surpassing the pre2009 baseline at all latitudes.

— here: add 95%

confidence intervals!



9. heat budget decomposition

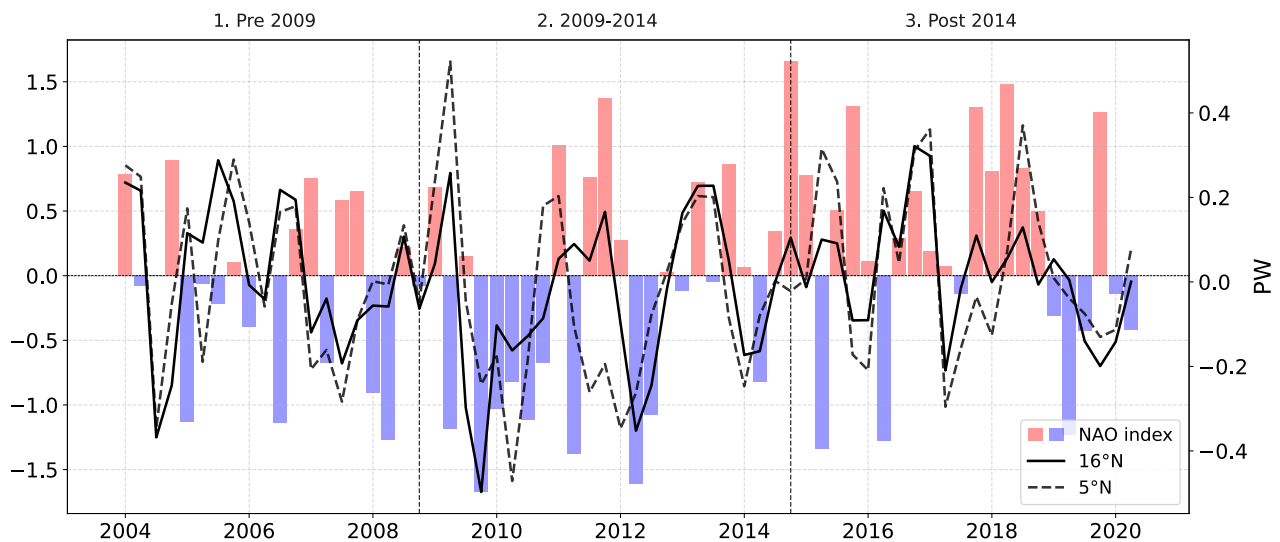
the reduction in MHT is due to ? Compared to the pre 2009 baseline the air sea heat flux in the region 16°N - 5°N increases around 15% while the heat transport convergence decreases by only 1.5%.

Maybe table for this

10. I. Found high correlation of the subtropical north atlantic heat transport with the NAO index for the period 2009-2014 of around 0.50 where the strong negative anomalies in MHT correlate with a negative NAO event in winter of 2009 and 2010, compared to the pre2009 baseline (-0.2),

Connection to coherence split?

(Plot: MHT anomalies, not detrended vs. NAO index)



South of 5°N there is no such correlation visible. Disconnected ? Gyre circulation (discussion)

1.2 Research Question

1. Are MHT variations propagating coherently across the atlantic?
2. Are there specific events and locations where they don't?
3. (*If so, what causes those breaks and differences?*) - depends a bit on the result of the heat budget stuff

1.3 Answer

1. MHT variations north of 35°N are not coherent with those south throughout the whole timeperiod. South of 35°N there are periods with high coherence throughout the atlantic until 35°S.
2. In the period of 2004-2008 latitudes from 26°N southward are connected with latitudes in the south atlantic up until 25°S. However there is no connection at 35°N with latitudes south of 16°N, aswell as no connection of 35°S with latitudes north of 5°N through the whole time period. The interconnection between 26°N until 25°S underwent a shift in 2009; with the subtropical region of the North Atlantic (35°N-16°N) and the South Atlantic region (south of 5° N) correlate within themselves, they are no longer linked to one another. This pattern of two covarying boxes stays persistent and continues through 2014. Thereafter, a recovery in the connection is found, resulting in an enhancement of the coherence, which is found to be superior to its previous state (two covarying boxes are 45°N-16°N & 16°N-35°S).
3. (*There is a high correlation at the break latitude 16°N with the NAO index, indicating this as a trigger for the coherence break. There is an overall reduction in the MHT across all latitudes which seems connected with an overall circulation reduction, as there is even an increased heat uptake from the atmosphere at the break region of 16°N-5°N which is not compensated by northward transport but instead the ocean heat content in the region increases compared to the baseline period 2004-2009.*)

1.4 Background

1. The ocean's role in heat uptake and distribution is essential, when considering the Earths Energy Imbalance, where the ocean is the dominant heat sink (91% of excess heat is absorbed by the oceans.) (*IPCC, 2021, WG I, Ch. 7*)
2. Once the heat enters the ocean it is either stored or recirculated, hence transported. (*Trenberth and Fasullo (2017)*). This conservation of heat leads to an equation for meridional heat transport.
3. Atlantic has a unique role in heat uptake and distribution, due to the AMOC all heat is transported northward, compared to other ocean basins (*Trenberth and Caron (2001)*) and accounts for around 88% of the total northward heat transport at 26°N.
4. Climate models predict possible AMOC weakening (*IPCC (2021), WG I, Chapter 9*), which would inturn influence the heat redistribution and uptake (*Impacts: (Marshall et al. 2013)*)
5. Need for observational datasets, not only consisting of in situ measurements but incorporating satellite observations, as those can account for longer timeseries and spatial resolution as an mooring array or argo data can account for.
6. The AMOC-MHT interconnection is well-established at 26°N, but due to limited spatial and temporal observation remains poorly understood on basin wide scales. (Interannual variability

driven by climate modes, AMOC/MHT drop in 2009 at 26°N, following record low NAO) (Bryden et al (2020))

1.5 Experimental Approach

1. Using the satellite based estimates from Calafat et al. (2025) I analyzed the coherence of meridional heat transport in the Atlantic at 12 latitude bands, spanning from 65°N to 35°S using statistical methods such as correlation and wavelet coherence.
2. This dataset also offers air-sea heat flux, heat transport convergence and ocean heat content tendency data, which I used to decompose the heat budget in the region of 16-5°N, where the coherence between subtropical and south Atlantic changes.

1.6 Implication(s) / Conclusion

The coherence break around 2009 was likely triggered by the anomalous negative NAO, which lead to a AMOC reduction of around 2.5 Sv in the period of 2009-2016.

This in turn lead to a reduction of MHT across all latitudes in my dataset. Where at the break latitudes (16-5°N) there as also an increased heat uptake with increased the ocean heat content in this region.

This matches with the finding:

„In the words of McCarthy et al. (2012), for each change there is “less overturning and more recirculation.”

.....

Events such as a record low nao are able to trigger changes in the AMOC along with the heat transport which persist up to 5 years for the MHT

However time mean MHT recovered after 2014 to an even higher transport across all latitudes.

??

Events such as the record low NAO in 2009 is one factor that lead to a reduction in AMOC (this is known from lit) accompanied by a reduction in MHT and a coherence break, possibly driven by gyre circulation, persistent for several years, however both are able restore its pre break state / strength afterwards.

Record low NAO -> AMOC weaker -> less overturning , more recirculation (McCarthy et al. (2012))
-> MHT reduction as AMOC reduction (Johns et al, amoc carries 88% of mht) -> coherence break between subtropical and south Atlantic (coordinates for gyres) since mht is carried within gyres by gyre circulation rather than being transported northwards due to the weaker overturning ! -> recovery of AMOC after (how many years, find lit for this?) -> recovery of both mean MHT and coherence (reconnected through 16°N-5°N)

(Answer) In this study I found that MHT tends to be coherent across the atlantic south of 40°N until 35°S (like kelly et al 2014 found), however in 2009 the coherence drops between subtropical north atlantic and south atlantic latitudes, along with an overall MHT reduction during a period of some years, with a following recovery. This event is likely linked to the record low NAO end of 2009

(Implication/Conclusion)

These results indicate that an event such as the record low NAO in 2009 is one factor that can lead to a reduction in AMOC (this is known from lit) accompanied by a reduction in MHT and a coherence break, possibly driven by gyre circulation, persistent for several years, however both are able restore its pre break state / strength afterwards.

Hypothesen- Paper

Background

Question

Experimental
Approach

Results

Answer

Implication

1.7 raw concept abstract

(Background)

The Atlantic ocean is one of the dominant heat sinks for excess heat in the climate system, with a high uptake and redistribution driven by the Atlantic Meridional Overturning Circulation (AMOC). Understanding how the Atlantic meridional heat transport (MHT) varies across latitudes and also connects with the AMOC is essential.

(mht-amoc connection: johns et al)

Understanding how the heat transport varies across latitudes and also interconntes with the AMOC is essential, when evaluating how global warming impact the heat content of the ocean.

(Question)

However, due to sparse observations the basin wide variability of MHT remains poorly understood, which leads me to investigate wether variations of MHT propagate coherent throughout the Atlantic basin, and if there are specific events which can trigger alterations.

(Exp approach/Methods) Using satellite-based estimates from Calafat et al. (2025) I evaluated the coherence of Atlantic MHT at 12 latitude bands between 2004-2020, spanning from 65°N to 35°S. I used statistical methods such as correlation and wavelet coherence and connecting MHT variations to the NAO index. *(Decomposition of heat budget?)*

(Result) I found that in 2004-2020 the MHT variations north of 35°N are disconnected to those south.

Moreover, in 2009 the relationship between the subtropical north Atlantic latitudes and the south Atlantic latitudes changed, leading to 2 covarying boxes which are no longer connected with each other, with a coherence drop specifically in the region of 16°N - 5°N.

In this period (2009-2014) MHT anomalies were more strongly correlated with the NAO index, than before and after.

However, after 2014 a recovery in the connection is found, resulting in an enhancement of the coherence as well as the mean heat transport values, which are both superior to its previous state.

(Answer / Implication / Conclusion)

These results indicate that an event such as the record low NAO in 2009 is one factor that can lead to a reduction AMOC (this is know form lit) which is accompanied by the found reduction of MHT (xy PW?) and a break in coherence across the identified break latitudes. However, the results also show that the heat transport, even if its change is persistent for several years, is able to restore is pre-break state afterwards even exceeding it.

Maybe link to gyre circulation is missing here? Just a bit duplication of the results instead of real implication?

1.7 nice concept abstract

(Background) The Atlantic ocean is one of the dominant heat sinks for excess heat in the climate system, with a high uptake and redistribution driven by the Atlantic Meridional Overturning Circulation (AMOC). Understanding how the Atlantic meridional heat transport (MHT) varies across latitudes and also connects with the AMOC is essential.

(Question) However, due to sparse observations the basin wide variability of MHT remains poorly understood, which leads me to investigate whether variations of MHT propagate coherent throughout the Atlantic basin, and if there are specific events which can trigger alterations.

(Exp approach/Methods) Using satellite-based estimates from Calafat et al. (2025) I evaluated the coherence of Atlantic MHT at 12 latitude bands between 2004-2020, spanning from 65°N to 35°S. I used statistical methods such as correlation and wavelet coherence and connecting MHT anomalies to the NAO index.

(Result) I found that in 2009 the relationship between the subtropical north Atlantic latitudes and the south Atlantic latitudes changed, leading to 2 disconnected (*covarying*) boxes, with a coherence drop specifically in the region of 16°N - 5°N. However, after 2014 a recovery in the connection is found, resulting in an enhancement of the coherence as well as the mean heat transport values, which are both superior to its previous state. In the weaker period (2009-2014) MHT anomalies were more strongly correlated with the NAO index, than before and after.

(Answer / Implication) These results indicate that an event such as the record low NAO in 2009 can affect not only the overall heat transport across the Atlantic but can change the coherence pattern locally. However, the results also show that the heat transport, even if its change is persistent for several years, is able to restore its pre-break state afterwards, even exceeding it.